Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
1	(a)		The [vector] sum of the momenta of bodies in a system stays constant [even if forces act between the bodies] accept overall momentum remains constant (1) provided there is no external / resultant force (1) accept closed system **Accept**: Total momentum of a system (or bodies) before a collision (or explosion) = total momentum after collision (explosion)... (1) ....provided no external / resultant forces act (1) accept closed system	2			2		
	(b)	(i)	Correct application of C of M e.g. Expression for momentum before collision: 0.20×9 [= 1.8] (1) Expression for momentum after collision: $(0.20 \times -2) + 0.40v$ (1) Convincing algebra: v = [+] 5.5 [m s^{-1}] (1) Error in 'signs' (e.g. for rebounding snowball gains no credit and **no ecf** for final answer) **Alternative:** Δp for snowball = -2.2 [N s] i.e. 0.2(-2 – (+9)) (1) Δp for hat = + 2.2 [N s] (1) **ecf** Hence $v_{\text{hat}} = \frac{2.2}{0.4}$ = 5.5 [m s^{-1}] (1)		3		3	3	
		(ii)	$\Delta p_{\text{hat}} = 5.5 \times 0.40 = 2.2$ [N s] **ecf** on v from (b)(i) (1) $F = \frac{2.2}{0.14}$ = 15.7 [N] (Accept 16 N) (1) **Alternative:** $a_{\text{hat}} = \frac{5.5}{0.14}$ = 39.3 [m s^{-2}] **ecf** on v from (b)(i) (1) $F = 0.4 \times 39.3 = 15.7$ [N] (1)		2		2	2	

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	(c)		Explanation: Momentum change (decrease) of snowball is reduced / less (compared to (b)(i)) (1) Momentum change (increase) of hat is reduced / less or compared with 2.2 [N s] (1) Therefore velocity (increase) of hat is less (than in (b)(i)) **and** so Natalie is incorrect (1) If calculation and conclusion only, award maximum 2 marks (treat as neutral): e.g. $1.8 = 0.6v$ (1) $v = 3$ [m s^{-1}] and hence Natalie is incorrect (1)			3	3		
			Question 1 total	**2**	**5**	**3**	**10**	**5**	**0**